<table>
<tr><th></th><th></th></tr>
<tr><td>1</td><td>Define an Operating System. Mention its main functions.</td></tr>
<tr><td>2</td><td>Distinguish between system software and application software</td></tr>
<tr><td>3</td><td>State inter- Process communication (IPC)</td></tr>
<tr><td>4</td><td>State fragmentation. Difference between internal and external fragmentation.</td></tr>
<tr><td>5</td><td>Differentiate between process and thread.</td></tr>
<tr><td>6</td><td>Explain OS structure with microkernel approach.</td></tr>
<tr><td>7</td><td>Differentiate between One-to-One multithreading models and many-to-many multithreading models with diagram</td></tr>
<tr><td>8</td><td>Describe page replacement algorithms
(FIFO,LRU,OPTIMAL)with examples</td></tr>
<tr><td>9</td><td>Elaborate Process Life Cycle with diagram</td></tr>
<tr><td>10</td><td>Solve the FCFS CPU Scheduling Algorithm
<table>
<tr><th>Process</th><th>CPU (Burst Time in Millisecond)</th></tr>
<tr><td>P1</td><td>5</td></tr>
<tr><td>P2</td><td>24</td></tr>
<tr><td>P3</td><td>16</td></tr>
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</table>
Calculate 1) Gantt chart 2)Average waiting time
3)Average Response time 4)Average turnaround time.</td></tr>
<tr><td>11</td><td>Discuss about contiguous Memory Allocation in detail</td></tr>
<tr><td>12</td><td>Explain Segmentation With Paging</td></tr>
<tr><td>13</td><td>Demonstrate the above terms in Details
1. Demand Paging
2. Swapping
3. Paging
4. Disk I/O</td></tr>
</table>

	5. Disk Cache 6. I/O Buffering
14	The job are scheduled for execution as follows. solve the problem by using i) SJF <table><tr><th>Process</th><th>Arrival</th><th>Burst Time</th></tr><tr><td>P1</td><td>0</td><td>8</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>9</td></tr><tr><td>P4</td><td>3</td><td>5</td></tr></table>
15	Describe the Evolution of Operating System in detail
16	Discuss the **structure of an Operating System** with a neat diagram.
17	Describe various **Operating System services**.
18	Compare **Batch OS, Multiprogrammed OS, and Time-sharing OS**.
19	Define a process and explain the **process life cycle** with diagram.
20	Demonstrate the **Inter-Process Communication (IPC)** mechanisms.
21	Explain the CPU Scheduling Algorithum write Advantages & Disadvantage.
22	Identify the **Critical Section Problem** and its requirements.
23	Apply the FCFS CPU Scheduling Algorithm <table><tr><th>Process</th><th>CPU (Burst Time in Millisecond)</th></tr><tr><td>P1</td><td>5</td></tr><tr><td>P2</td><td>24</td></tr><tr><td>P3</td><td>16</td></tr><tr><td>P4</td><td>10</td></tr><tr><td>P5</td><td>3</td></tr></table>Calculate 1) Gantt chart 2)Average waiting time 3)Average Response time 4)Average turnaround time.

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<tr><td>25</td><td>Justify Segmentation with diagrams.</td></tr>
<tr><td>26</td><td>Identify Virtual Memory and Demand Paging.</td></tr>
<tr><td>27</td><td>Explain Page Replacement Algorithms: FIFO and LRU.</td></tr>
<tr><td>28</td><td>Prepare File allocation methods and free-space management.</td></tr>
<tr><td>29</td><td>Manage Disk scheduling algorithms and disk management.</td></tr>
<tr><td>30</td><td>Explain OS security, Virtual Machines, and Cloud OS concepts.</td></tr>
<tr><td>31</td><td>Compare Batch OS, Multiprogrammed OS, and Time-sharing OS.</td></tr>
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<tr><td>41</td><td>Describe threads and multithreading models and its types with diagram</td></tr>
<tr><td>42</td><td>Differentiate between demand paging and page replacement algorithms (FIFO,LRU,OPTIMAL)with examples</td></tr>
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<tr><td>44</td><td>Discuss Disk scheduling algorithms and disk management.</td></tr>
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<tr><td>49</td><td>Expresses various Operating System services</td></tr>
<tr><td>50</td><td>What is Segmentation?</td></tr>
<tr><td>51</td><td>Differentiate between process & thread.</td></tr>
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ii) SJF
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53	Elaborate the CPU Scheduling Algorithum write Advantages & Disadvantage.
54	Prepare File allocation methods and free-space management
55	Describe demand paging and page replacement algorithms (FIFO,LRU,OPTIMAL)with examples
56	Demonstrate the above terms in Details 5. Swapping 6. Paging 7. Disk I/O 8. Disk Cache 9. I/O Buffering
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58	Examine directory implementation method and file allocation method (linked, Indexed, contiguous)
59	Explain OS security, Virtual Machines, and Cloud OS concepts
60	Demonstrate Deadlock prevention and deadlock recovery techniques.